



G H Raisoni Institute of Engineering and Management, Pune.

(An Autonomous Institution Affiliated to SPPU)

S. Y. B. Tech. (AI & AI-ML)

Question Bank

Subject: Computer Architecture and Organization

1. What are the different types of instruction are used in microprocessor?
2. What are the different addressing modes are used in Instruction set architecture?
Explain any two by giving an example.
3. What are the logical instructions used in microprocess? Explain any two of them.
4. What are the arithmetic instructions used in microprocess? Explain any two of them.
5. Compare the CISC and RISC Instruction set architecture.
6. Explain instruction execution cycle.
7. What are the branching instructions used in microprocess? Explain any two of them.
8. Solve the following for fixed point arithmetic.
 - i. $14 + 9$
 - ii. $18 - 7$
 - iii. 13×11
 - iv. $213 / 5$
9. What are the branching instructions used in microprocess? Explain any two of them.
10. List different types of Register used in microprocess and explain the purpose of any three registers.
11. Explain the Instruction execution cycle with the help of diagram.
12. What are the different addressing modes are used in instruction set architecture?
Explain any two addressing mode with the help of example.

13. Explain following instructions:

- i. MOV A, R3
- ii. ADD A, R1
- iii. SUBB A, @R0
- iv. ANL A, 50H
- v. XRL A, #22H

14. Explain rotate instruction of microprocessor.

15. Differentiate between RISC and CISC architecture.

16. What are the logical instructions used in microprocess 8085? Explain any two of them.

17. What is Sub-Word parallelism?

18. What are the arithmetic instructions used in microprocess 8085? Explain any two of them.

19. Explain Carry Look Ahead adder.

20. What are the branching instructions used in microprocess 8085? Explain any two of them.

21. What are the branching instructions used in microprocess? Explain any two of them.

22. What is IEEE standard for single precision format?

23. List different types of Register used in microprocess 8085 and explain the purpose of any three registers.

24. Explain basic function block of a computer.

25. Do the following fixed point operation using binary number system

- i. 8+12
- ii. 14-6
- iii. 5*7
- iv. 15/4

26. Perform the following Floating point operation using single precision IEEE format.

- i. 8.25+9.5
- ii. 12.75-8.25
- iii. 3.5*1.375
- iv. 12.0/2.875

27. What is sub-word parallelism?

28. Explain Booth algorithm for Multiplication.
29. Explain Booth algorithm for Division.
30. Explain IEEE Single Precision format for floating point number.
31. Explain IEEE Double Precision format for floating point number.